

CAUSALITY ASSESSMENT OF AN ADVERSE DRUG REACTION

Internship Case Study – Pharmacovigilance

Case ID	TASK3-AMX-001
Patient	45-year-old female
Suspected drug	Amoxicillin
Adverse reaction	Skin rash
Onset	3 days after starting amoxicillin
Outcome	Recovered after discontinuation of the suspected drug

1. Case Summary

A 45-year-old female patient developed a skin rash approximately three days after initiation of amoxicillin. The reaction was temporally associated with the suspected medicine and improved after the medicine was stopped. No positive rechallenge is documented in this training case.

Relevant Case Data

- The suspected medicine is amoxicillin, an antibacterial agent.
- The reported adverse event is a skin rash.
- The time-to-onset is approximately 3 days after treatment initiation.
- Improvement followed withdrawal of the suspected drug (positive dechallenge).
- Rechallenge information is not available; therefore, rechallenge should not be assumed.
- The case is designed for pharmacovigilance causality-assessment practice.

2. Objective

To assess whether a causal relationship exists between amoxicillin exposure and the reported skin rash, using standard causality-assessment principles and to classify the relationship as Certain, Probable/Likely, or Possible where appropriate.

Important: This is an educational case for internship/practical training. It does not represent a real patient record and should not be used to make clinical decisions.

CAUSALITY ASSESSMENT

3. Assessment of Key Causality Criteria

Criterion	Case finding	Interpretation
Temporal relationship	Rash occurred 3 days after amoxicillin was started.	Compatible temporal sequence; supports a causal relationship.
Dechallenge	Rash improved/recovered after stopping the suspected drug.	Positive dechallenge; supports causality.
Rechallenge	No rechallenge is documented.	Cannot be used to strengthen the assessment.
Alternative causes	The case provides no documented alternative cause or competing medicine.	Alternative explanations are not established from the available data.
Known pharmacological/clinical association	Skin rash is a recognized adverse reaction reported with amoxicillin and related penicillins.	Background compatibility supports, but does not prove, causality.

A Naranjo-style checklist can be used as a structured aid. The exact score depends on the completeness and wording of the case record; the following scoring is an educational example based only on the information supplied in this case.

Question	Response	Example points
Previous conclusive reports on this reaction?	Yes / known association	+1
Event after drug was administered?	Yes	+2
Improved after withdrawal?	Yes	+1
Reappeared on readministration?	Not done / unknown	0
Alternative causes identified?	Not established in case	+2
Objective evidence?	Not provided	0
Illustrative total		6

Interpretation of the illustrative score: A score in the 5–8 range is generally interpreted as Probable in the Naranjo algorithm. The score here is only a training illustration; the final classification should follow the assessment method required by the internship and the complete case information.

5. Data Gaps to Check in a Real PV Case

- Exact dose, route, formulation, and treatment indication.
- Other medicines, recent infections, foods, and relevant medical history.
- Description, severity, distribution, and timing of the rash.
- Laboratory/diagnostic findings, if relevant.
- Exact date of discontinuation and date of recovery.
- Whether the reaction recurred on accidental or deliberate re-exposure.

FINAL CLASSIFICATION & INTERNSHIP RECORD

6. Causality Conclusion

Based on the information provided, the case has a compatible temporal relationship, a positive dechallenge, and no documented rechallenge. No competing cause is established in the supplied case. These findings support a causal association between amoxicillin and the reported skin rash.

Classification	Meaning in this case
Certain	Would require stronger evidence such as a convincing temporal relationship, recovery on withdrawal, and usually a positive rechallenge or very strong evidence excluding alternatives. The present case does not document rechallenge.
Probable / Likely	Supported by a reasonable time relationship and improvement after withdrawal, with no stronger alternative explanation documented. This is the most appropriate training classification for the supplied information.
Possible	Would be considered when timing is compatible but alternative causes or insufficient information make the relationship less convincing.

Final training classification: PROBABLE / LIKELY

Rationale: compatible onset after exposure + improvement after withdrawal + no documented rechallenge + no alternative cause established in the supplied case.

7. Pharmacovigilance Documentation Checklist

Item	Status for this training case
Patient demographics	Available
Suspected drug	Available – amoxicillin
Adverse event	Available – skin rash
Time to onset	Available – 3 days
Dechallenge	Positive / recovery reported
Rechallenge	Not documented
Alternative causes	Not established from supplied data
Causality category	Probable / Likely
Case limitations	Limited clinical detail; educational scenario

Internship submission note

You can use this document as the supporting PDF for the causality-assessment task shown in your screenshot. If your internship portal asks for a GitHub link, upload this PDF to your repository and submit the repository/file link.

